

AC-APP-ELLA APP USING FLUTTER AND FIREBASE

Mini Project Report

*Submitted to the APJ Abdul Kalam Technological University in
Partial fulfilment of requirement for the award of degree*

Bachelor of Technology

In

Computer Science

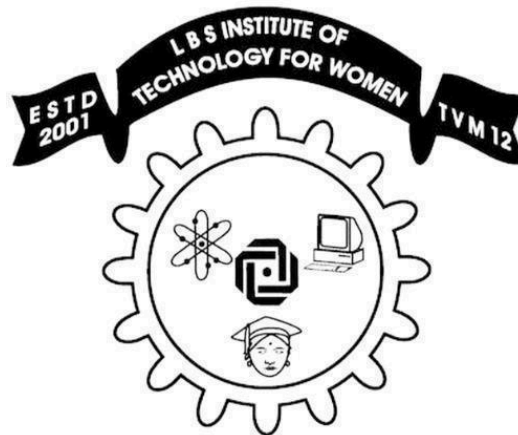
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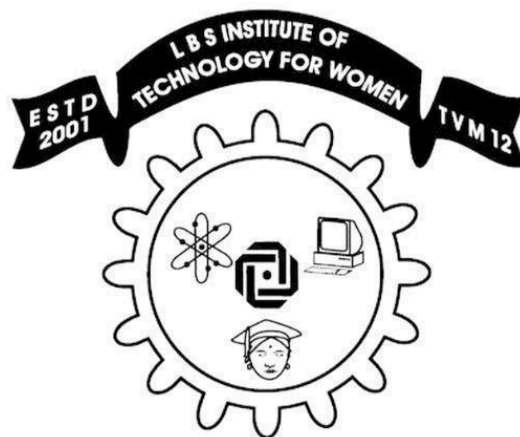
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LBS INSTITUTE OF TECHNOLOGY FOR WOMENS

TRIVANDRUM, KERALA

APRIL 2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

This is to certify that the report entitled “ **AC-APP-ELLA** ” submitted by **KEERTHANA MANOJ (LBT22CS064), NANDANA RAJESH KP (LBT22CS078), LEKSHMI R NAIR (LBT22CS068), PANCHAMI M (LBT22CS091)** to the APJ Abdul Kalam Technological University in partial fulfilment of the award Of the requirement for the award of degree of Bachelor of Technology in Computer Science And Engineering is a Bonafide record of the mini project carried out by them under our guidance and supervision.

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DECLARATION

We undersigned hereby declare that the project report **AC-APP-ELLA APP**, submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a Bonafide work done by us under the supervision of Dr Dileep V K, Associate Professor, Department of Computer Science Engineering, LBS Institute of Technology for Women ,Poojappura. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misinterpreted or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University

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ABSTRACT

AC-APP-ALLA Club App is a mobile application developed using Flutter and Firebase to streamline and modernize college club management. Taking inspiration from platforms like the TinkerHub app, AC-APP-ALLA offers a refined and user-friendly interface tailored specifically to address the common pain points faced by student clubs and organizations. On many campuses, clubs operate independently with little synchronization, resulting in important event announcements getting lost in a flood of information and scattered communication platforms. As a result, students often miss out on events they would have otherwise been interested in. A more critical issue arises when students register for events without actually attending, yet continue to claim benefits such as duty leave and certificates, leading to administrative challenges and unfair advantages. AC-APP-ALLA tackles these inefficiencies by incorporating key functionalities such as secure student authentication, smooth event creation and registration, and an intelligent attendance management system using QR code scanning. The app ensures that only verified attendees are acknowledged for participation, promoting fairness and transparency. The login system is role-based, distinguishing between club members and committee members, with added support for Google Sign-In to streamline access. With Firebase as the backend, the app benefits from real-time updates, secure data handling, and scalable performance. Flutter's flexible and expressive UI framework enhances the visual appeal and responsiveness across devices, resulting in a cohesive user experience. Through this project, we aim to demonstrate how modern app development tools can be leveraged to build a digital ecosystem that enhances communication, boosts collaboration, and simplifies the operational workflow of student-led organizations. AC-APP-ALLA not only reduces manual overhead and eliminates inconsistencies but also lays the foundation for future enhancements such as notifications, analytics, and personalized user features to further elevate student engagement and club productivity.

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CHAPTER 1

INTRODUCTION

In colleges and universities, student clubs play an essential role in promoting extracurricular involvement, skill development, and leadership among students. These clubs organize various activities such as workshops, seminars, competitions, and social events that contribute to the holistic development of students beyond academics. However, the efficient management of these clubs is often hindered by outdated and manual methods. Organizing events, keeping track of attendance, managing participant records, and communicating updates to members can be time-consuming and error-prone when done using spreadsheets, paper-based records, or informal channels like social media groups. Such traditional approaches result in several challenges — event mismanagement due to lack of coordination, inaccurate or manipulated attendance records, difficulty in maintaining a central record of activities, and ineffective communication among club members. With increasing digital dependency and the need for systematic organization, there is a strong demand for a digital solution that can handle these operational issues seamlessly and promote smoother functioning of clubs. Inspired by the structure and community-driven approach of TinkerHub, a tech community known for its vibrant club activities and organized processes, AC-APP-ELLA was conceptualized as a smart solution to bring similar digital efficiency into our own college club management system. AC-APP-ELLA is a Flutter-based mobile application specifically designed to meet the administrative and operational needs of college student clubs. The application offers a centralized platform where all stakeholders — organizers, committee members, and general participants — can interact based on their assigned roles. Users can log in using a secure, role-based authentication system, which ensures that they can access and manage only the features relevant to their position in the club. Organizers can create, edit, and manage events, while committee members can oversee all ongoing activities, approve participation, and track progress. Club members can browse upcoming events, register for activities, and view their own participation history. A standout feature of the app is its QR code-based attendance system, which replaces manual sign-ins with a secure and efficient method of tracking attendance.

The app provides a personalized dashboard that displays real-time information about events, announcements, and user-specific data. Built using Flutter, AC-APP-ELLA works seamlessly across Android, iOS, and web platforms. Firebase Firestore is used as the backend service, enabling real-time data updates, cloud storage, and user authentication with minimal latency. By digitizing the core operations of student club management, AC-APP-ELLA simplifies administrative processes and improves the overall user experience. It reduces dependency on physical records and external tools, thus minimizing errors and data redundancy. With real-time updates and instant data synchronization via Firebase, the app ensures transparency and consistent communication among all stakeholders. The use of QR code-based attendance further adds a layer of security and accuracy, eliminating the risk of proxy attendance and manual errors. AC-APP-ELLA promotes greater engagement by allowing students to view events easily and participate without barriers. For organizers, the burden of manual coordination is significantly reduced, allowing them to focus more on planning meaningful activities rather than handling logistics. In the long term, this application can be scaled to support multiple clubs across institutions, with features like notifications, analytics dashboards, and feedback collection. It represents a step toward modernizing campus life and empowering student communities with smart technology solutions. Overall, AC-APP-ELLA is not just an app—it's a complete ecosystem designed to revolutionize college club management in the digital age. The development of AC-APP-ELLA followed a structured, modular approach to ensure scalability, security, and ease of maintenance. The project began with requirement analysis by identifying the specific needs of student clubs, such as event creation, user roles, attendance management, and communication gaps. Based on these insights, a feature list was defined, and the application flow was designed using wireframes and low-fidelity prototypes. The frontend of AC-APP-ELLA was developed using Flutter, a UI toolkit by Google that enables the creation of natively compiled applications for mobile, web, and desktop from a single codebase. Flutter was chosen for its expressive UI, fast development cycle, and strong community support. The backend services were built on Firebase, a cloud-based platform also by Google, which offers robust features like Firebase Authentication for secure login, Cloud Firestore for real-time NoSQL database services, and Firebase Storage for uploading and managing media files like QR codes. The implementation includes clearly separated modules for user authentication, role management, event handling,

and QR code generation and scanning. Role-based access control ensures that organizers, committee members, and participants can access only the features relevant to their role. Events are stored in Firestore collections and dynamically updated, allowing users to view real-time changes. QR code-based attendance tracking was implemented using a Flutter package that generates QR codes for events and scans them securely during the check-in process. Testing was performed on multiple devices and platforms to ensure a smooth and responsive user experience. Feedback from peers and mentors was integrated to refine the interface and improve usability. The result is a well-structured, responsive, and secure application that reflects both technical depth and practical utility in real-world scenarios.

1.1 PROBLEM STATEMENTS

College clubs face significant operational challenges due to their reliance on outdated and fragmented management systems. Currently, clubs depend on a disconnected mix of communication channels like WhatsApp groups, paper-based attendance registers, and standalone Google Forms for registrations. This inefficient approach creates three major pain points: (1) poor event visibility as activities get lost across scattered platforms, leading to low participation despite organizers' efforts; (2) administrative inefficiency with time-consuming manual processes that remain vulnerable to errors and attendance fraud; and (3) lack of centralized coordination making it difficult to manage multiple clubs or track campus-wide engagement. The AC-APP-ELLA CLUB APP addresses these issues through an integrated digital solution featuring username/password authentication, event listing and QR-code attendance verification. Built with Flutter and Firebase, this platform eliminates the limitations of manual systems while preventing fraudulent participation, ultimately transforming how colleges manage club activities and student engagement in the digital era

1.2 OBJECTIVE

To design and develop a comprehensive mobile application that modernizes and streamlines the management of college student clubs by offering a centralized, secure,

and user-friendly digital platform to enhance operational efficiency, student engagement, and administrative transparency.

1.2.1 SUB-OBJECTIVES

- To digitize and unify all aspects of club management, replacing fragmented manual and digital systems with a single cohesive platform.
- To improve student engagement by increasing the visibility, accessibility, and ease of participation in club events and activities.
- To implement secure QR code-based attendance tracking, ensuring fair participation and eliminating fraudulent claims related to event attendance.
- To minimize administrative workload for organizers by automating repetitive processes such as event registration, attendance logging, and member tracking.
- To create a connected campus environment by facilitating seamless interaction between students, club committees, and administrators.
- To uphold data privacy and security through robust authentication systems and role-based access control mechanisms.
- To ensure the scalability and long-term viability of the system, enabling support for multiple clubs and a growing user base without compromising performance.
- To promote transparency in club operations through features like real-time data synchronization, activity logs, and event analytics.
- To ensure inclusivity and ease of use for all users by offering a responsive and intuitive Flutter-based UI design adaptable to different devices and user skill levels.
- To equip institutions with a future-ready tool that evolves alongside emerging digital trends and student expectations.

1.3 SUMMARY

AC-APP-ELLA is a club management application developed to streamline the organization and participation processes of student-run clubs in colleges. The app was inspired by the effective practices followed by TinkerHub, a tech community known for its structured and impactful club activities. Traditional methods of club management including handwritten attendance sheets, informal social media announcements, and scattered event records often lead to miscommunication, poor coordination, and data loss. AC-APP-ELLA addresses these challenges by providing a centralized digital platform where club activities can be managed efficiently by organizers, committee members, and participants. The application features a role-based login system, ensuring that users access features specific to their responsibilities. Organizers can create and manage events, committee members can oversee activities and verify details, and participants can register for events and view their own participation history. One of the core innovations of the app is its QR code-based attendance system, which replaces manual sign-ins with a more secure and accurate method. A personalized dashboard provides users with real-time updates on events, past participation, and relevant announcements, enhancing transparency and engagement.

Built using Flutter, AC-APP-ELLA runs smoothly across Android, iOS, and web platforms. The backend uses Firebase Firestore, enabling real-time data synchronization, secure user authentication, and cloud-based storage of records. This ensures that any updates made by organizers or committee members are instantly reflected across all users' devices, improving communication and operational flow. By digitizing the end-to-end club management process, AC-APP-ELLA minimizes administrative burden, increases student involvement, and creates a scalable, user-friendly environment for managing campus events. The paper discusses the development of an NSS club mobile app for GIT using Android Studio. It focuses on improving communication and management with features like live updates, chat, profile management, activity tracking, and event management. The app uses Firebase for backend services and Google Sheets for data management administrative burden, increases student involvement, and creates a scalable, user-friendly environment for managing campus events.

CHAPTER 2

LITERATURE REVIEW

1. S. Faniband , A. Kumbar, D. Naik, V. Nikam and P. Shrivastsa, "Development of Mobile Application Using Android Studio for NSS Club of GIT,"inResearchGate, May 2024 [1].

The research paper *"Development of Mobile Application Using Android Studio for NSS Club of GIT"* presents a well-structured project aimed at enhancing communication and coordination within the NSS club at GIT. The paper effectively identifies the inefficiencies in traditional methods, such as fragmented communication and lack of centralized event management, justifying the need for a dedicated mobile app. The proposed solution leverages Android Studio for development, Firebase for backend services, and Google Sheets for data management, offering features like real-time updates, chat functionality, and activity tracking. The methodology follows a systematic approach, from requirements analysis to deployment, supported by use case and sequence diagrams. However, the paper could benefit from deeper technical insights into implementation details, performance metrics, and scalability considerations, particularly regarding the use of Google Sheets for larger datasets. Despite its strengths, the paper lacks empirical data or user feedback to validate the app's real-world impact post-deployment. Additionally, security measures and compliance with data privacy regulations are not discussed, which are critical for institutional adoption. The literature review is robust, referencing relevant studies on Firebase and Android development, but the absence of comparative analysis with existing solutions weakens its academic rigor. Overall, the project is a commendable effort that addresses a genuine need, but future iterations could focus on scalability, security, and user validation to further solidify its contributions to digital club management in educational settings.

2. P. Pabba, B. Viswadha, P. Srivani, P. Rajashree and S. V. Kumar, "Mobile Application for College Event Management," in International Research Journal of Engineering and Technology (IRJET), vol. 9, no. 12, pp. 201-207, Dec. 2022 [2].

The paper "Mobile Application for College Event Management" presents a Flutter and Firebase-based solution called "VNR CONNECT" designed to streamline event and club management at VNR Vignana Jyothi Institute of Engineering and Technology. The authors effectively identify the limitations of manual processes and propose a centralized digital platform featuring event registration, club management, and real-time notifications. While the paper demonstrates a clear understanding of the problem space and offers a technically sound solution using modern frameworks, it lacks empirical validation through user testing or performance metrics. The literature review is comprehensive, covering 30+ relevant works, but the discussion would benefit from deeper technical details about the implementation, particularly regarding data security and scalability considerations. The proposed application shows promise for improving student engagement and administrative efficiency, though future work should address security measures, include user feedback mechanisms, and provide comparative analysis with similar systems to strengthen its academic contribution and practical applicability. The paper serves as a valuable reference for institutions considering digital transformation of campus event management systems.

3. K. Om, P. Lokalwar, P. Pawar, A. Khan, and K. M. Kharde, "College Department Mobile Application by Using Flutter," *International Research Journal of Modernization in Engineering, Technology and Science (IRJMETS)*, vol. 6, no. 5, pp. 7713–7715, May 2024 [3].

The paper "College Department Mobile Application by Using Flutter" presents a Flutter-based mobile app designed to improve communication and workflow management in academic departments. While the authors effectively identify institutional challenges like inefficient collaboration and propose a centralized solution with features like instant messaging and task management, the study lacks empirical validation through user testing or performance metrics. The technical implementation, though leveraging Flutter's cross-platform advantages, lacks depth in discussing critical aspects like state management, API integration, or security measures. The paper's strength lies in its practical approach to digitizing academic workflows, but its claims of improved efficiency remain unsubstantiated without quantitative data. Future work should include rigorous user studies, security enhancements, and comparative analysis with existing solutions to strengthen its academic and practical contributions. The study serves as a basic reference for Flutter applications in education but requires more robust evidence to demonstrate its impact.

4. R. Deshmukh, V. Rajbhar, M. Sankhe, and R. K. Kahlon, "Android Application for College Events," *International Research Journal of Engineering and Technology (IRJET)*, vol. 7, no. 9, pp. 3467–3471, Sep. 2020 [4].

The paper "Android Application for College Events" presents a Flutter and Firebase-based mobile solution designed to streamline event communication in academic institutions by replacing traditional notice boards with real-time digital notifications. The authors effectively identify key challenges in current college communication systems, such as delayed message dissemination and low student engagement, proposing a centralized platform featuring role-based event posting, push notifications, and bookmarking capabilities. While the technical implementation leveraging Firebase's scalability and Flutter's cross-platform functionality is well-explained, the study lacks empirical validation through user adoption metrics or performance benchmarks. The comparison table highlighting improvements over legacy systems (e.g., internet-based vs. Bluetooth notifications) strengthens the paper's practical relevance, but critical aspects like data security measures and integration with existing campus management systems remain unaddressed. The project demonstrates a clear understanding of user needs through features like theme customization and Google authentication, though its actual impact on student participation would benefit from quantitative analysis. This work serves as a valuable case study for institutions adopting mobile notification systems, but future iterations should incorporate user feedback, security protocols, and broader institutional integrations to enhance its academic contribution and real-world applicability.

5. L. Bhanot, R. Shyam, M. T. Khan, P. Bahadur, and S. Ali, "EVENTO - An Android App for Event Planners," *International Research Journal of Modernization in Engineering, Technology and Science (IRJMETS)*, vol. 6, no. 4, pp. 6669–6676, Apr. 2024 [5].

The paper "EVENTO - An Android App for Event Planners" presents a mobile application designed to streamline event management through features like task coordination, vendor communication, and expense tracking, addressing key challenges in the field. While the study effectively outlines the app's user-friendly interface and comprehensive functionalities (including note-saving and invoicing tools), it lacks empirical validation through user testing or performance metrics, limiting the assessment of its real-world efficacy. The proposed system shows promise for improving event planning efficiency but would benefit from technical

details on backend architecture, security measures, and comparative analysis with existing solutions. Future enhancements could include AI integration and cross-platform compatibility to broaden its applicability, though the current research serves as a solid conceptual foundation for event management technology.

6. S. A. Bhagat, S. G. Dudhalkar, P. D. Kelapure, A. S. Kokare, and S. A. Bachwani, "Review on Mobile Application Development Based on Flutter Platform," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 10, no. 1, pp. 804–809, Jan. 2022 [6].

The paper titled “*Review on Mobile Application Development Based on Flutter Platform*” by S. A. Bhagat et al., published in the *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, volume 10, issue 1, January 2022, presents an in-depth analysis of Flutter as a framework for mobile application development. This review is particularly valuable for students, developers, and professionals exploring modern tools for building cross-platform applications. The authors begin by explaining the fundamental architecture of Flutter, emphasizing its widget-based approach, which allows developers to build highly customizable UIs using pre-built or custom widgets. They further describe how the Dart programming language, used by Flutter, contributes to improved performance, readability, and ease of development. Features like hot reload, which enables developers to view code changes in real-time without restarting the application, are highlighted as key productivity boosters. The review also delves into Flutter’s layered architecture, covering how rendering, layout, and event handling are separated efficiently. It discusses state management techniques such as Provider and BLoC, which are essential for maintaining application logic and user interface consistency. Additionally, the paper touches upon testing methodologies provided by Flutter for unit testing, widget testing, and integration testing—tools that are important for maintaining code quality and reliability during development.

2.1 TABLE OF LITERATURE REVIEW

SL N O	TITLE	METHODOLOGY	PROS	CONS
1.	S. Faniband , A. Kumbar, D. Naik, V. Nikam and P. Shrivastsa, "Development of Mobile Application Using Android Studio for NSS Club of GIT,"inResearch Gate,May2024.	The paper discusses the development of an NSS club mobile app for GIT using Android Studio. It focuses on improving communication and management with features like live updates, chat, profile management, activity tracking, and event management. The app uses Firebase for backend services and Google Sheets for data management	1.Enhanced Communication & Coordination 2.User-Friendly 3.Scalable 4.Efficient Event Management	1.Limited Platform Availability 2.Potential Data Security Issues 3.Dependency on Internet Connectivity
2.	P. Pabba, B. Viswadha, P. Srivani, P. Rajashree and S. V. Kumar, "Mobile Application for College Event Management," in International Research Journal of Engineering and Technology (IRJET), vol. 9, no. 12, pp. 201-207, Dec. 2022	Proposes a user-friendly platform with Firebase backend for efficient event coordination, real-time updates, and secure data management, aimed at reducing manual errors and enhancing reliability	1.SimplifiesCoordination 2.User-Friendly 3.Interface Secure & Reliable	1.Limited to one college 2. No advanced analytics 3.Needs internet

3.	K. Om, P. Lokalwar, P. Pawar, A. Khan, and K. M. Kharde, "College Department Mobile Application by Using Flutter," <i>International Research Journal of Modernization in Engineering, Technology and Science (IRJMETS)</i> , vol. 6, no. 5, pp. 7713–7715, May 2024.	The document discusses the development of a college department mobile application using Flutter. The app aims to improve internal communication, workflow, and collaboration by providing features like instant messaging, information sharing, and task management. The study highlights the app's positive impact on efficiency and suggests future improvements, such as performance monitoring and additional customizations.	1. Cross-Platform Compatibility 2. Improved Communication & Collaboration 3. User-Friendly Interface	1. Data Management Challenges 2. Integration Issues 3. Performance Optimization
4.	R. Deshmukh, V. Rajbhar, M. Sankhe, and R. K. Kahlon, "Android Application for College Events," <i>International Research Journal of Engineering and Technology (IRJET)</i> , vol. 7, no. 9, pp. 3467–3471, Sep. 2020	This mobile app streamlines college event communication using Firebase and Flutter. It offers real time push notifications, a virtual events board, urgent alerts, and bookmarking. With secure Google login, it enhances student engagement and replaces outdated .	1. Real-Time Notifications 2. flutter and firebase is used 3. Enhanced User Experience 4. Secure and Reliable	1. Notification Fatigue 2. Network Dependency 3. Limited Personalization

5.	L. Bhanot, R. Shyam, M. T. Khan, P. Bahadur, and S. Ali, "EVENTO - An Android App for Event Planners," <i>International Research Journal of Modernization in Engineering, Technology and Science (IRJMETs)</i> , vol. 6, no. 4, pp. 6669–6676, Apr. 2024.	"EVENTO" is an Android app that simplifies event planning by improving task coordination, financial management, and communication. It features a user friendly interface, task tracking, note-saving, and expense management for seamless collaboration. Future enhancements may include AI driven recommendations and third-party integrations for greater efficiency.	1.Enhanced Event Organization 2.User-Friendly Interface 3.Financial Transparency	1.Compatibility Issues 2.Market Competition 3.Security & Privacy Concerns
6.	S. A. Bhagat, S. G. Dudhalkar, P. D. Kelapure, A. S. Kokare, and S. A. Bachwani, "Review on Mobile Application Development Based on Flutter Platform," <i>International Journal for Research in Applied Science & Engineering Technology (IJRASET)</i> , vol. 10, no. 1, pp. 804–809, Jan. 2022.	The document reviews Flutter, a cross-platform framework for iOS and Android app development using a single Dart-based codebase. It covers Flutter's architecture, state management with GetX, Firebase integration, and testing methods.	1.Single code base . 2.Hot reload 3.High performance 4.Rich ui & widgets Firebase Integration	1.Large App Size 2. Limited third party libraries 3.Dart learning curve 4.iOS-Specific Challenges

CHAPTER 3

REQUIREMENT SPECIFICATION AND DESIGN

3.1 SOFTWARE REQUIREMENTS

AC-APP-ELLA is developed using the Flutter framework for cross-platform UI and Firebase as the backend service provider. The software requires a stable internet connection, access to a modern web browser or Android/iOS device, and Firebase project credentials for data storage and authentication. Development tools such as Android Studio or VS Code are used for coding and debugging the application.

- **Flutter Framework**

- Flutter is the primary framework used to develop the AC-APP-ELLA application. It allows the creation of a single codebase that runs seamlessly on Android, iOS, and the web. Its widget-based architecture ensures a responsive and visually appealing user interface.

- **Dart Programming Language**

- Dart is the programming language used in Flutter for developing the application. It supports object-oriented programming, asynchronous operations, and Just-In-Time (JIT) compilation for smooth development, as well as Ahead-Of-Time (AOT) compilation for optimized performance in production.

- **Firebase Firestore Database**

- Firestore is a cloud-based NoSQL database that stores event details, user profiles, and attendance records in real time. It provides fast and scalable data retrieval with offline access, ensuring smooth app functionality even without an internet connection.

- **Integrated Development Environments (IDEs)**

- The application is developed using Android Studio and Visual Studio Code. Android Studio provides advanced debugging tools, an emulator for testing, and built-in Flutter support, while VS Code offers a lightweight and efficient coding environment with essential Flutter and Firebase extensions.

3.2 HARDWARE REQUIREMENTS

- Processor: AMD Ryzen 5 5500U with Radeon Graphics
- RAM: 8 GB
- Operating System: Windows 11

3.3 FUNCTIONAL REQUIREMENTS

The system must allow users to log in securely using their role-based credentials and access functionalities according to their role—organizer, committee member, or participant. Organizers can create events, committee members can manage attendance and approve participants, and users can view and register for events. The application must also support QR code generation and scanning for attendance marking.

- **User Registration and Login**

Users, including committee members and club members, can sign up using a registration form to create their accounts. They can log in using a username and password after sign-up, ensuring a simple and accessible login process.

- **Home Screen Navigation**

After logging in, committee members are directed to the Home Screen Committee, while club members are taken to the Home Screen Club Member. These home screens serve as a hub for their respective functionalities, allowing committee members to manage events and club members to explore and register for events.

- **Execom Tab Access**

Both committee members and club members have access to the Execom Tab, which provides important details about the executive committee's activities. This tab may contain information about club leadership, roles, and responsibilities within the organization.

- **Event Management**

Committee members can manage events through the Events Tab, where they can add, edit, view, delete, and scan event details. They can define event descriptions, specify dates and locations, and handle participant registrations efficiently.

- **Event Registration**

Club members can browse available events listed in the Events Tab and register for the ones they are interested in attending. This feature allows members to stay informed about upcoming activities and actively participate in club events.

- **QR-Based Attendance Tracking**

Each event generates a unique QR code that participants must scan to mark their attendance. This system ensures a streamlined and fraud-proof attendance process, reducing manual errors and ensuring accurate event participation records.

- **Database Management**

Firestore serves as the central database for storing event details, user profiles, and attendance records. It enables seamless data storage, retrieval, and synchronization, ensuring all users have access to the latest information in real time.

- **Authentication System**

The authentication system verifies users through email and password login, ensuring secure access to the app. This method helps restrict access to authorized users only, preventing unauthorized use and maintaining data privacy.

3.4 NON-FUNCTIONAL REQUIREMENT

The application should be responsive, user-friendly, and accessible across multiple devices and platforms including Android, iOS, and web. It must ensure data consistency and real-time updates using Firebase, with minimal loading time and reliable performance under normal usage. Security measures such as encrypted authentication and restricted role-based access must be implemented to protect user data and app integrity.

- **Reliability:** Ensure the app operates reliably with minimal downtime. Maintain consistency in real-time data syncing and notifications.

- **Scalability:** Design the app to handle increased users, events, and data as the number of clubs grows.
- **Security:** Protect user data, including personal details and event information, using secure Firebase authentication and database practices. Implement access control to restrict unauthorized actions.
- **Usability:** Provide an intuitive user interface for both students and admins. Ensure the app is easy to navigate, reducing the learning curve.
- **Performance:** Optimize the app for fast loading times that ensure smooth transitions between screens.
- **Maintainability:** Use modular code with proper documentation.
- **Adaptability:** Ensure the app can be easily extended to include new features, like notification types or integration with other systems.
- **Data Backup and Recovery:** Implement automated data backup in Firebase to prevent data loss in case of system failures.

3.5 OVERALL ARCHITECTURE

The architecture of the AC-APP-ELLA club management application is designed to provide a structured and secure platform for managing club activities. The system distinguishes between two primary user roles: Committee Members and Club Members, each with tailored functionalities and access privileges. The user journey begins with either a "Sign-up" process or a "Sign-in" for existing users. The "Sign-up" pathway directs users to role-specific Registration Forms, collecting necessary information and securely storing their credentials (Email/Password) within the Firestore users database. This user data is managed and authenticated by the Firebase Authentication service, ensuring secure access control. For returning users, the "Sign-in" process requires entering their registered Email/Password, which is

then verified against the Authentication service. Upon successful authentication, users are directed to role-specific home screens. Committee Members access the Home Screen Committee, which provides a gateway to administrative functionalities through an Execom Tab. Navigating through this tab leads them to the Events Tab, where they can perform crucial event management tasks. These include Add/Edit/View/Delete event details and utilize the Scan functionality for the QR code-based attendance system. All event-related data is persistently stored and managed within the Firestore Events database. Club Members, upon logging in, are directed to the Home Screen Club Member. From this interface, they can primarily view/Register for upcoming events, potentially also accessing event details through an Execom Tab available to them. The QR code element likely plays a role in event attendance tracking, either for member identification or for scanning by committee members.

The underlying data layer consists of two main Firestore collections: Firestore Events for storing event details and Firestore users for managing user profiles and authentication information. The Firebase Authentication service acts as a central point for verifying user identities and enforcing security policies across the application. The entire system is designed to facilitate efficient event management by committee members and seamless participation and information access for club members, all while ensuring data integrity and secure user authentication.

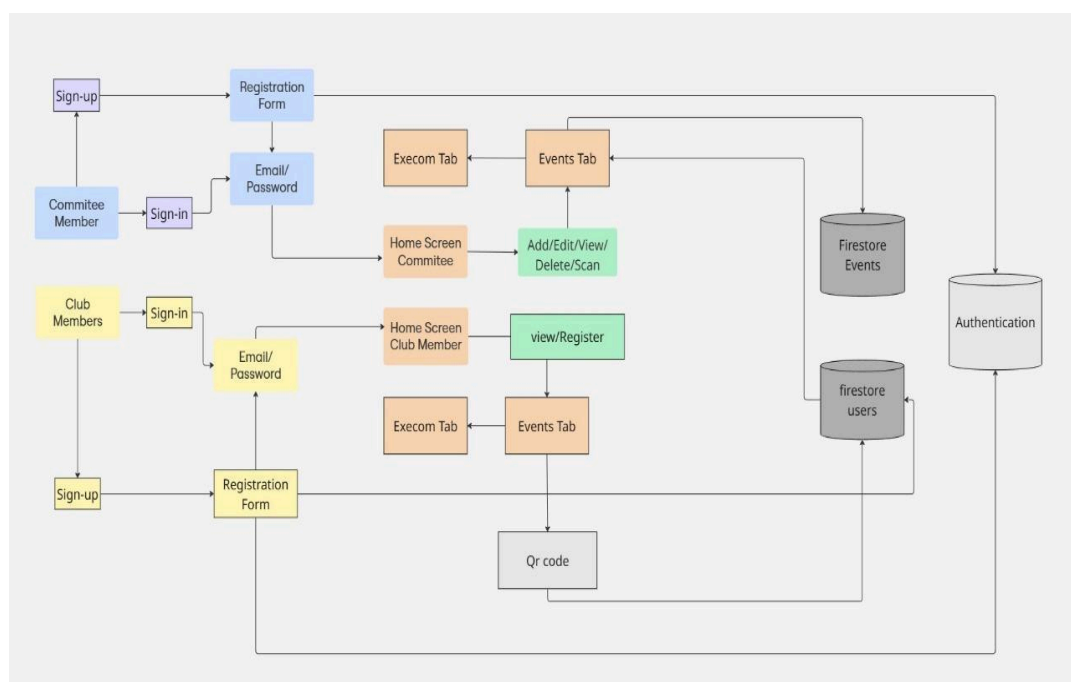


Fig 3.4 Overall Architecture

3.6 SYSTEM DESIGN

AC-APP-ELLA's system design centers on two distinct user roles with tiered functionalities. For Club Users, the platform provides essential capabilities including account registration/login and access to a personalized home page displaying relevant club information. Committee Users receive expanded administrative access through their dedicated interface, enabling them to manage critical club operations. The system features robust event management tools that allow both user types to view upcoming events while restricting editing/update privileges exclusively to Committee Users. A built-in review mechanism facilitates event registration and feedback submission, complemented by QR-code-based attendance tracking to ensure accurate participation records. This dual-role architecture creates an efficient hierarchy that balances user accessibility with necessary administrative controls, optimizing the club management process through clear permission delineation and streamlined workflows.

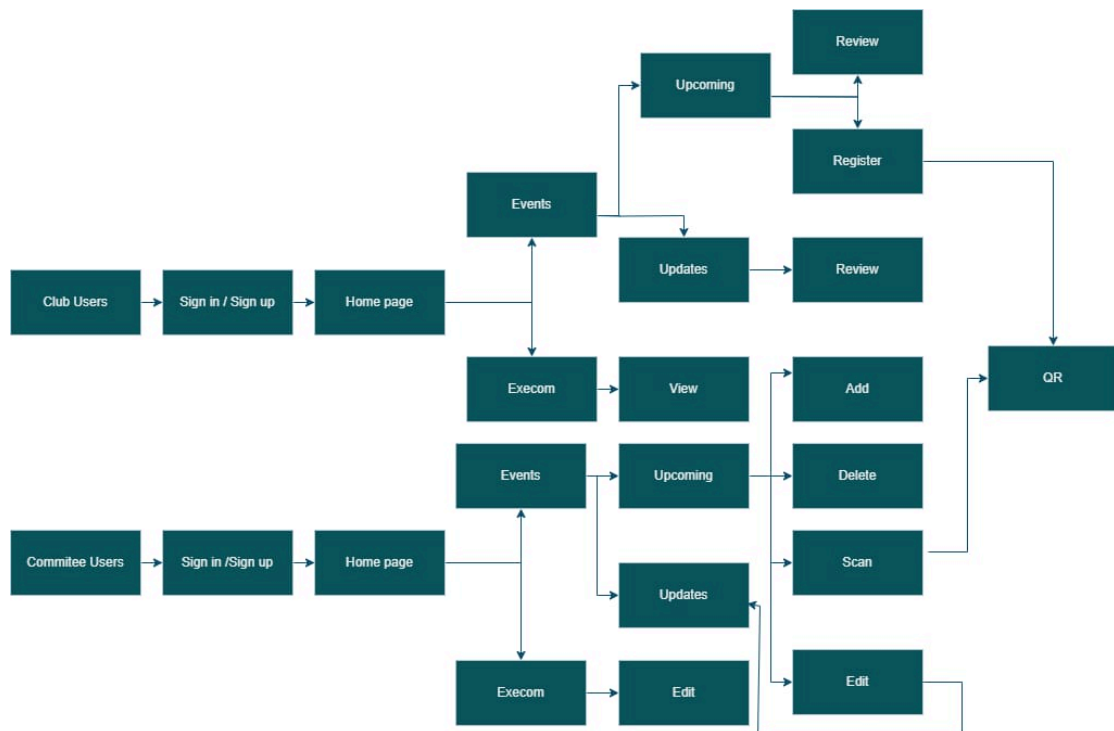


Fig 3.5 System Design

CHAPTER 4

METHODOLOGY

The development of AC-APP-ELLA followed a structured and phased methodology, integrating modern mobile development practices with a deep understanding of the user requirements and technical constraints associated with club management systems. The methodology consists of six major stages: Requirement Gathering, Design, Frontend Development (Flutter), Backend Development (Firebase), Integration, and Testing. Each phase was executed to ensure the creation of a scalable, secure, and user-friendly application.

4.1 REQUIREMENT GATHERING

The first step in the development process involved identifying and analyzing the target users—namely Committee Members and Club Members—and defining their specific roles and responsibilities within the app. Committee Members are responsible for creating and managing events, verifying attendance, and overseeing club operations, while Club Members interact with the platform primarily to view, register, and attend events. To accommodate the distinct responsibilities of these two user types, the app required a secure role-based access system. Essential features were determined through stakeholder discussions and competitor analysis. These features included event management capabilities, QR code-based attendance tracking, registration workflows, and secure authentication. Firebase Firestore was selected as the backend database due to its cloud-based infrastructure, support for real-time updates, and secure user data storage. A key requirement was the implementation of username and password authentication over third-party login services, offering more controlled and private access. During this phase, a constraint was introduced to prevent misuse of roles, as both Committee Members and Club Members utilize password-based authentication. To reduce the risk of role impersonation, passwords of Committee Members were required to start with the string "AC". If a login attempt is made with a Committee Member's username but a password not beginning with "AC," the app displays a message prompting the user to switch to the correct login type.

4.2 DESIGN

With requirements clearly defined, the design phase began by researching user expectations for event management applications and analyzing popular club management systems for inspiration. This research informed the structural planning of the application, ensuring that functionalities specific to Committee Members and Club Members were clearly separated and easy to access. Wireframes were created for the app's key screens, including login, registration, event listings, event creation, QR code generation and scanning, and event history tracking. A consistent and clean visual style was developed to support intuitive navigation and a pleasant user experience. The design was tailored for responsiveness, ensuring usability across different device types and screen sizes. Usability testing was conducted using interactive prototypes. Based on participant feedback, refinements were made to improve screen flow, reduce cognitive load, and enhance accessibility. This iterative approach ensured the design was both practical and aesthetically pleasing.

4.3 FRONTEND DEVELOPMENT(FLUTTER)

The Flutter development environment was set up and configured with necessary dependencies to begin building the app interface. Flutter's cross-platform capabilities allowed development of a unified codebase that works efficiently on Android, iOS, and web browsers. UI components were constructed using Flutter widgets to support functionalities like event listing, QR code scanning, and event management. User authentication was implemented using username and password systems. Special care was taken to apply the constraint for Committee Members' passwords, ensuring that login attempts validate the password format and guide the user appropriately. This mechanism added a layer of security without introducing complex role verification protocols. The event management system was developed to provide full CRUD (Create, Read, Update, Delete) functionality for Committee Members, while limiting Club Members to viewing and registering for events. The navigation between screens was optimized to maintain context and fluidity, enhancing the overall user experience.

4.4 BACKEND DEVELOPMENT(FIREBASE)

Firebase Firestore served as the backend solution for data management. Collections were created for users, events, and attendance records. Firebase Authentication was integrated to enable secure username-password logins, supported by role-based access rules. Real-time database synchronization allowed any updates by Committee Members to be instantly reflected in the app. Access controls were implemented based on the user role. Committee Members could create and modify event data, while Club Members were permitted to only view and register. Firebase security rules were established to prevent unauthorized access or data tampering. The QR code attendance system was built to record attendance in real-time. When a Club Member scans a QR code at an event, the system logs the participation with a timestamp, updating the event record in Firestore. Secure handling of this process was ensured to prevent duplicate or fraudulent check-ins.

4.5 INTEGRATION

The integration phase focused on connecting the frontend UI with the backend database services. Flutter's built-in support for Firebase facilitated a smooth connection for real-time updates and seamless data exchange. Event creation, editing, registration, and attendance marking were all synchronized using Firebase listeners and update triggers. The QR code scanning feature was directly linked to event data, ensuring that scanned entries are immediately reflected in the relevant attendance record. Data validation procedures were set in place to guarantee consistency across user actions and database entries. Additionally, the role-based login system with the "AC" password prefix for Committee Members was tested within the integration flow to ensure proper handling of user identity checks across authentication and UI access restrictions.

4.6 TESTING

A rigorous testing phase was conducted to ensure the app's functionality, stability, and usability. Unit tests were written for core components such as event creation, QR code generation and scanning, and user login validation. Each module was independently tested to verify expected behavior and performance. Integration testing was carried out to evaluate the interaction between the Flutter frontend and Firebase backend.

Scenarios such as real-time data updates, secure access control, and QR attendance tracking were simulated to check consistency and reliability. User testing sessions were held with both Committee Members and Club Members. Feedback was collected on usability, responsiveness, and feature clarity. Issues identified were debugged and resolved to deliver a smooth and reliable experience upon deployment. Throughout the development cycle, emphasis was placed on creating a secure and scalable platform. The design decision to include the "AC" prefix in Committee Member passwords served not only as a user verification technique but also as a lightweight method to mitigate potential misuse without adding complexity to the system architecture.

4.7 USE-CASE DIAGRAM OF AC-APP-ELLA APP

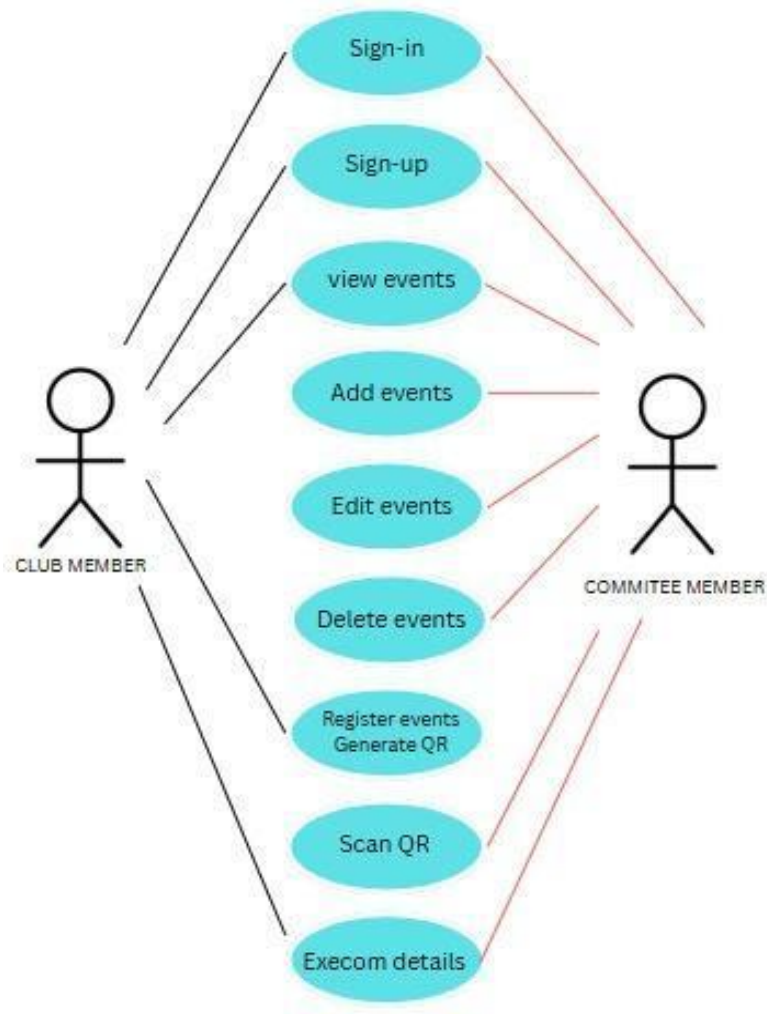


Fig 4.7 Use Case Diagram

CHAPTER 5

RESULT AND DISCUSSION

The development and implementation of AC-APP-ELLA have successfully optimized club management by streamlining event organization, attendance tracking, and user accessibility. The application provides a well-structured platform that enables Committee Members and Club Members to efficiently manage and participate in events. The username-password authentication system ensures a secure and user-friendly login process, allowing seamless access to the application without requiring external authentication services. One of the core functionalities of the app is event management, which allows Committee Members to create, edit, and delete events as needed. This feature ensures that clubs can efficiently schedule and organize activities, reducing the administrative workload while enhancing user engagement. The event details, including date, time, and location, are stored in Firebase Firestore, enabling real-time data synchronization and allowing users to access the most updated event information at any time. The QR code-based attendance tracking system provides an innovative and efficient method for marking attendance. Participants can scan a unique event QR code to register their presence, ensuring accuracy and preventing fraudulent attendance records. This system eliminates manual attendance tracking, making the process faster, more reliable, and less prone to errors. The scanned attendance records are instantly updated in the database, allowing Committee Members to view participation statistics in real time.

The dashboard feature serves as a central hub for users, displaying upcoming events, past event records, and other essential information relevant to the user's role. This enhances accessibility and usability, ensuring that users can quickly find the information they need. The intuitive interface ensures a smooth navigation experience, making it easy for users to interact with different features of the application. Data storage and management are handled efficiently using Firebase Firestore, which ensures that event details, attendance records, and user information are securely stored and updated in real-time. This enables seamless interaction between the frontend and backend, ensuring that all users have access to the most accurate and up-to-date information. The integration of Flutter and Firebase allows for

a scalable and high-performance system that can handle multiple users efficiently. Overall, AC-APP-ELLA has proven to be an effective solution for club management, simplifying event coordination, attendance tracking, and data management. The combination of a user-friendly interface, real-time data updates, and seamless functionality enhances the efficiency of club operations while providing an engaging experience for users. By leveraging modern technology, the app ensures that club activities are well-organized, accessible, and efficiently managed, contributing to improved participation & overall club management.

5.1 SCREENSHOTS OF AC-APP-ELLA APP

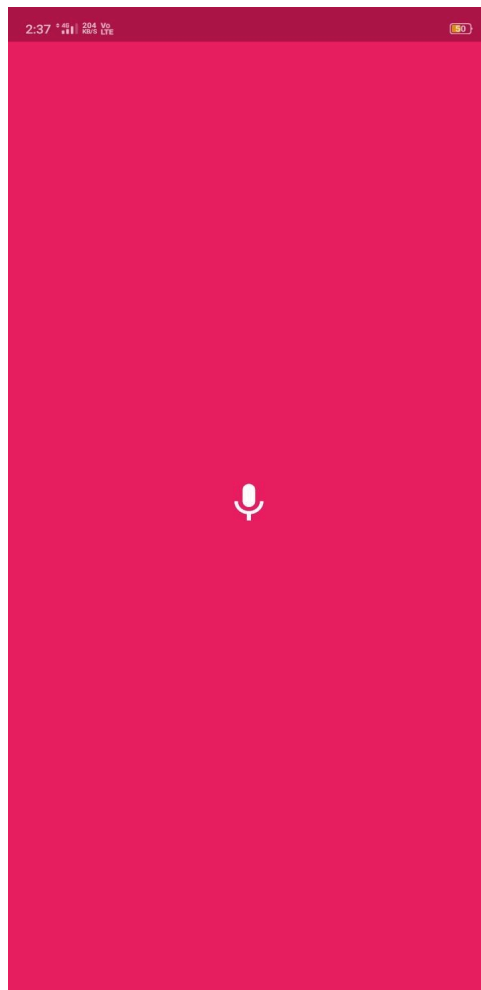


Fig 5.1 Splash Screen

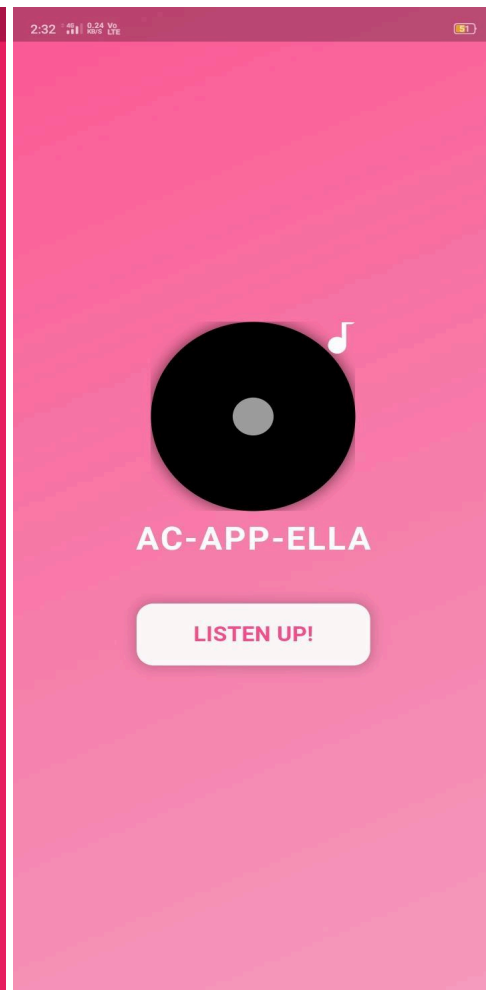


Fig 5.2 App logo screen

- The splash screen of the AC-APP-ELLA application features a vibrant pink background with a centrally positioned white microphone icon. This minimalist design likely serves as a brief visual introduction while the app initializes its core functionalities in the background.
- The Flutter-created logo screen for AC-APP-ELLA features a prominent black circular graphic with a central grey dot, reminiscent of a vinyl record. Positioned above the app's name in bold white letters, "AC-APP-ELLA," and a call-to-action button labeled "LISTEN UP!" below, the design suggests an audio or music-related theme for the application

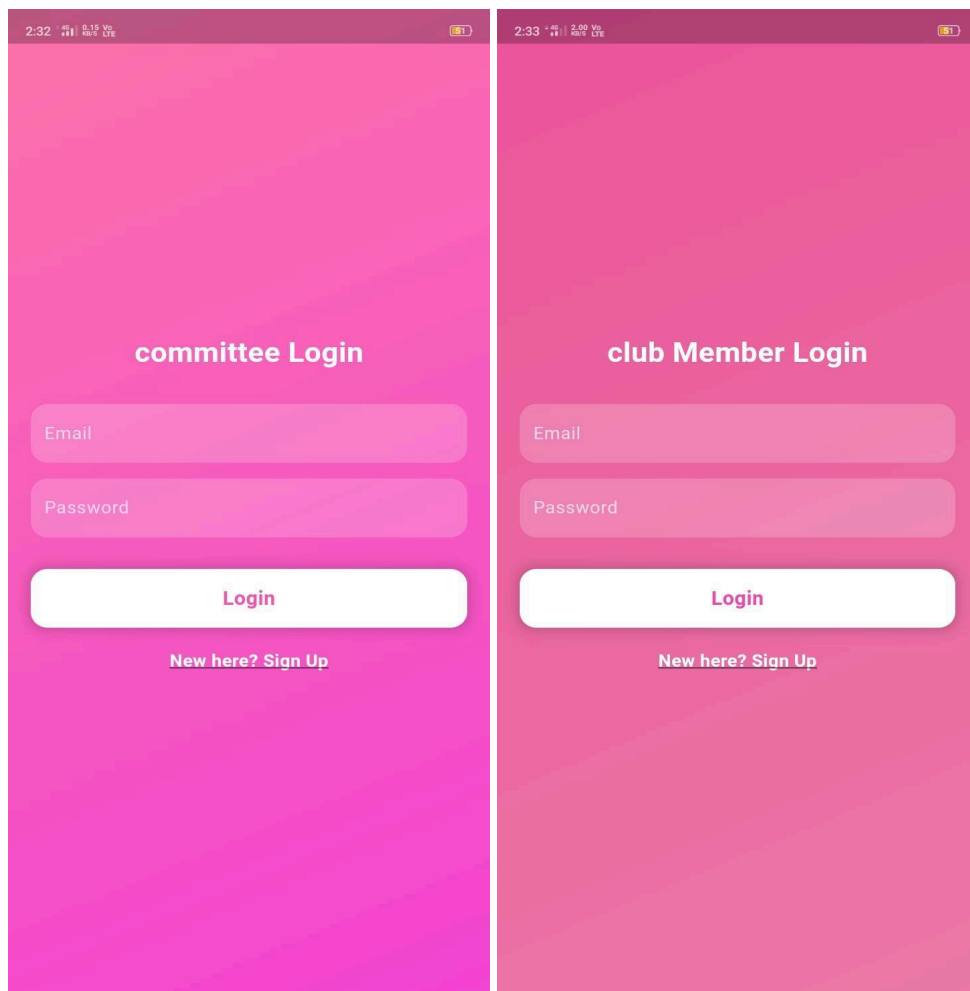


Fig 5.3 Committee Member Login Screen Fig 5.4 Club Member Login Screen

- This Flutter screen presents the "Committee Login" interface with a clean and straightforward design. It includes input fields for "Email" and "Password,"

accompanied by a prominent white "Login" button and a clear "New here? Sign Up" link for new committee members.

- This Flutter screen provides the login interface specifically for "Club Member Login," mirroring the design of the committee login screen for consistency. It features input fields for "Email" and "Password," a clearly visible white "Login" button, and a "New here? Sign Up" link for new club members to register.

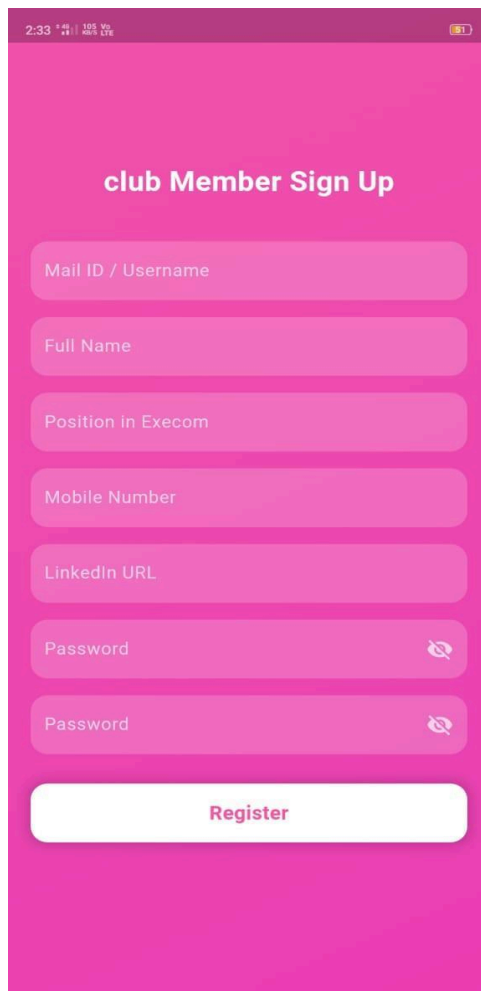


fig 5.5 Club Member Signup Screen

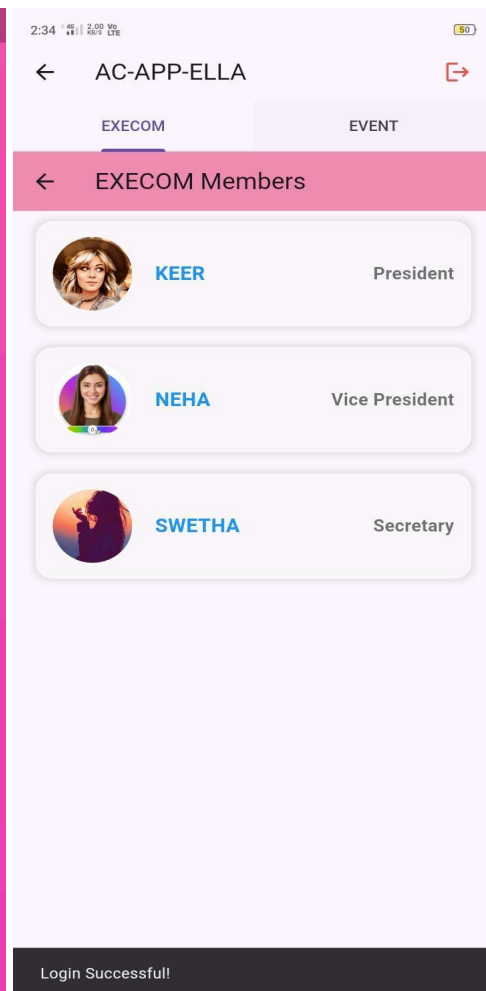


fig 5.6 Execom Tab For Both User

- This Flutter screen presents the "Club Member Sign Up" form, requiring new members to fill in details such as "Mail ID / Username," "Full Name," "Position in Execom," "Mobile Number," "LinkedIn URL," and to create and

confirm their "Password." A prominent white "Register" button at the bottom allows users to submit their information and create a new account.

- This Flutter screen displays a list of executive committee members, each represented with a profile picture, their name, and their designated leadership role within the club. The interface offers a straightforward way to view the organizational structure and identify the individuals in charge of different responsibilities.

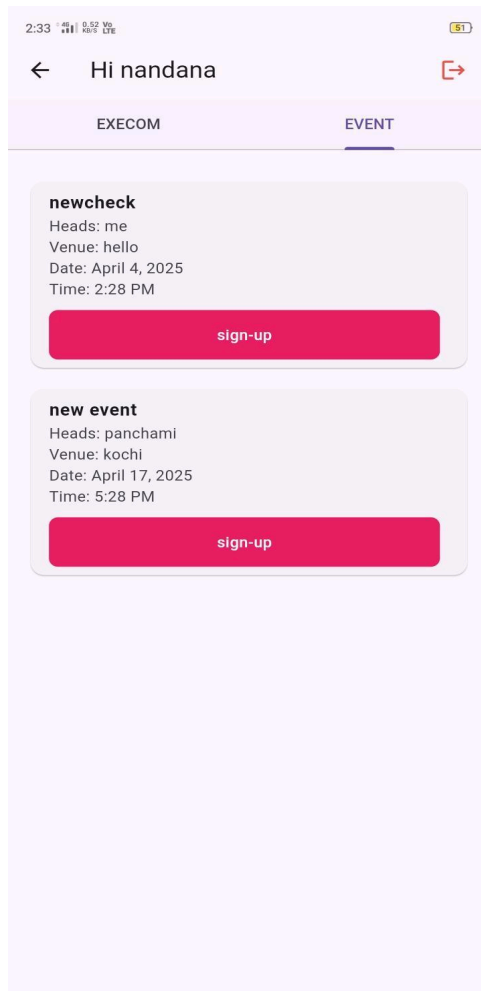


Fig 5.7 Club Member Home Screen
Events Tab

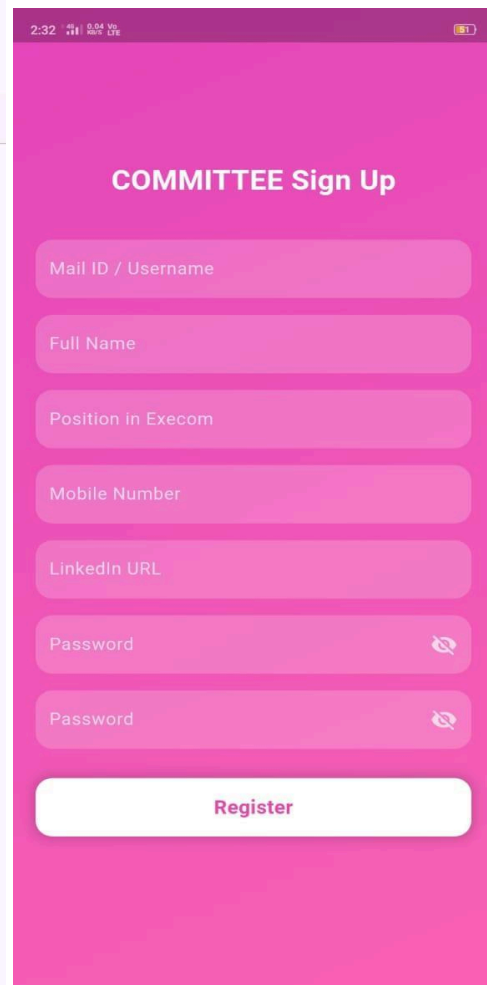


Fig 5.8 Committee Member Signup
Screen

- This Flutter screen displays a list of upcoming events under the "EVENT" tab. Each event card provides essential details such as the event title, the head organizer, venue, date , and time, along with a prominent "sign-up" button for easy registration.

- This Flutter screen presents the "COMMITTEE Sign Up" form, requiring prospective committee members to input information like their username/email, full name, position within the executive committee, mobile number, LinkedIn URL, and to create and confirm a password. A clear white "Register" button at the bottom allows users to submit their details to join the committee.

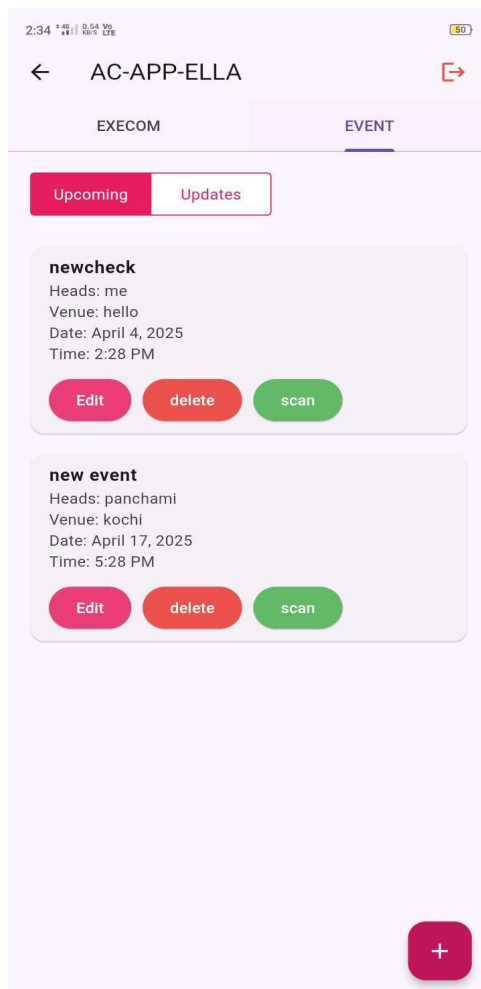


Fig 5.9 Committee Member Home
Screen

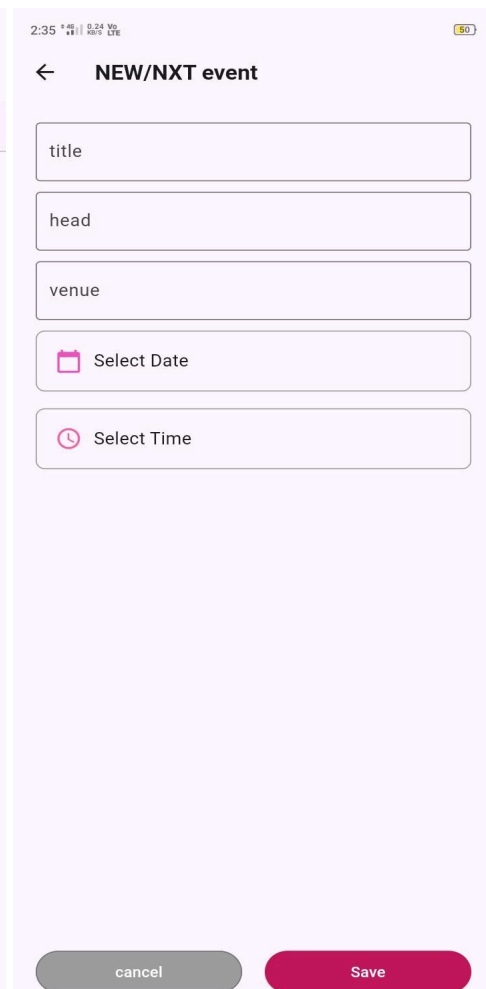


Fig 5.10 Add Event Screen

- This Flutter screen, under the "EVENT" tab for committee members, displays a list of events with administrative controls. For each event, such as "newcheck" and "new event," committee members have options to "Edit,"

"delete," and "scan" (likely for QR code attendance), indicating their ability to manage and oversee these activities.

- This Flutter screen provides a form to create a "NEW/NXT event," featuring input fields for the event "title," "head" organizer, and "venue." It also includes interactive elements to "Select Date" and "Select Time" for the event, along with "cancel" and "Save" buttons at the bottom for managing the creation process.

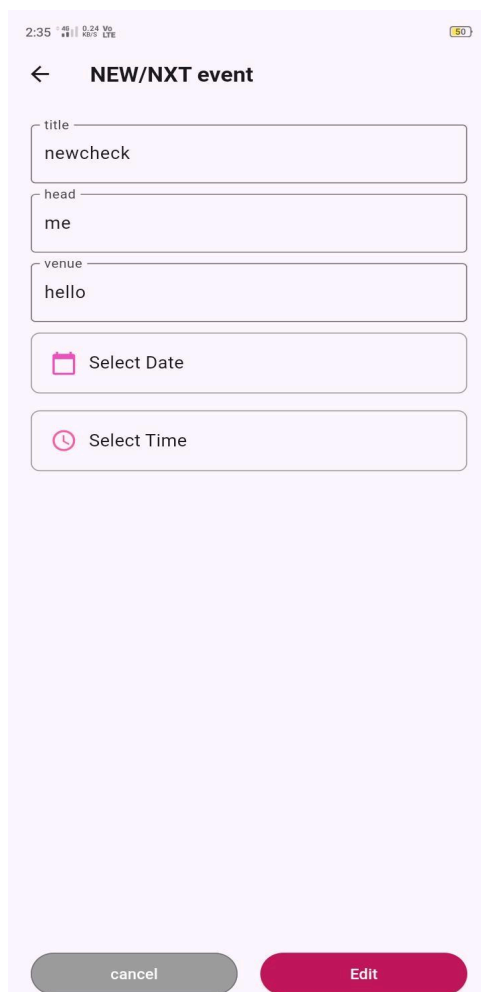


Fig 5.11 Edit Event Screen

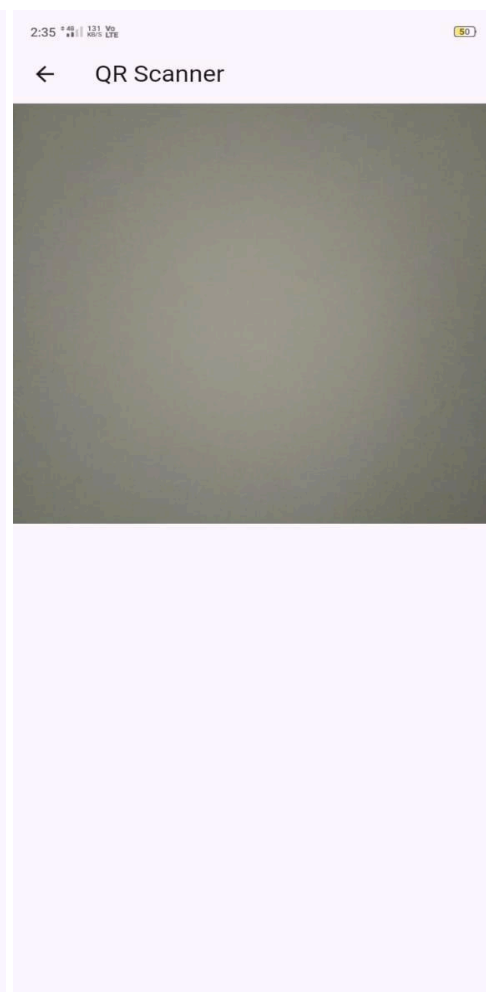


Fig 5.12 QR Scan

- This Flutter screen displays the "NEW/NXT event" creation form with some fields already populated, showing "newcheck" as the title, "me" as the head organizer, and "hello" as the venue. The options to "Select Date" and "Select Time" are still available, and the "Save" button has now changed to "Edit," suggesting an existing event is being modified.

- This Flutter screen presents a "QR Scanner" interface, likely utilizing the device's camera to scan QR codes for various functionalities within the app, such as attendance tracking for events. The prominent viewfinder area indicates where the QR code should be positioned for scanning.

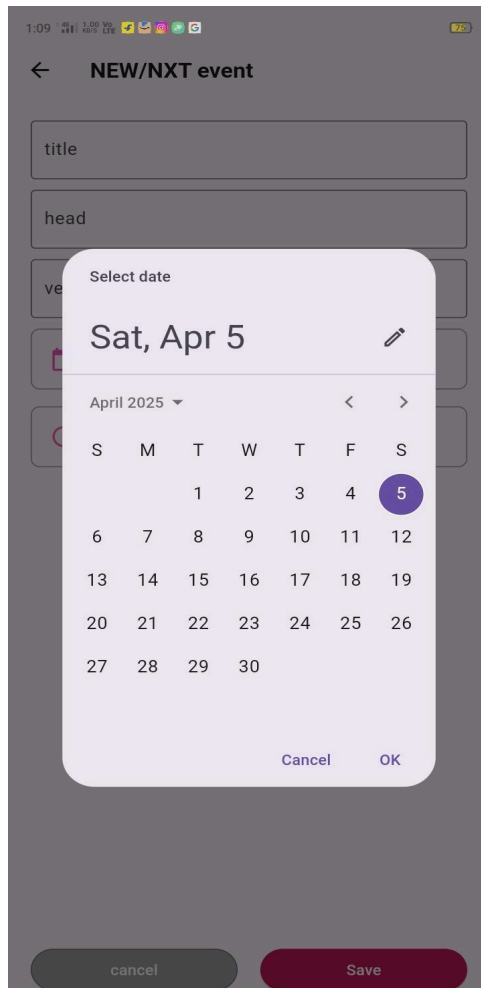


Fig 5.13 Add Event Screen -
Date Selection

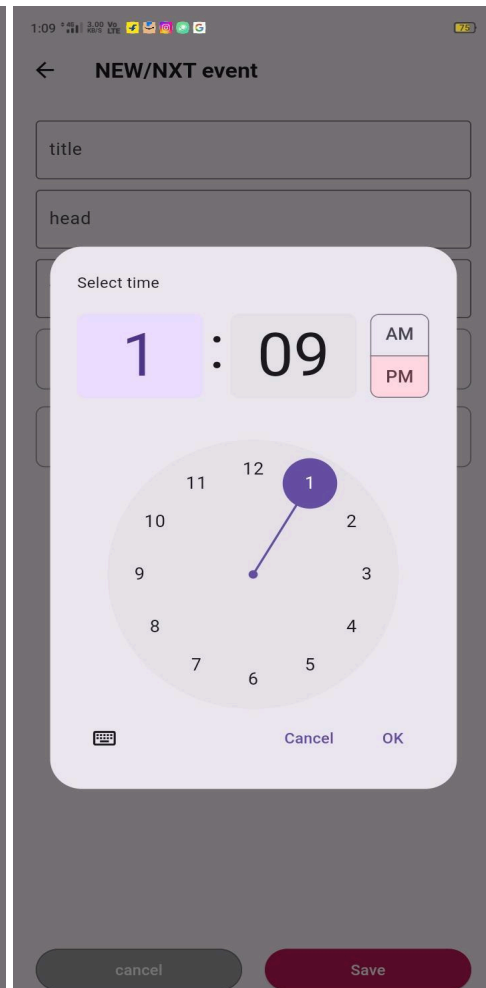


Fig 5.14 Add Event Screen -
Time Selection

- The "NEW/NXT event" creation screen now features an overlay for "Select date," presenting a calendar interface for users to pick the desired date for their event. This interactive calendar allows for easy navigation through months and days, providing a user-friendly way to schedule the event. Once a date is chosen, users can confirm their selection to update the event details.

- The "NEW/NXT event" creation screen now displays a "Select time" modal overlay, featuring a visual clock interface for setting the event's time. Users can manipulate the hour and minute hands on the clock or use the digital input

at the top, with AM/PM options available to specify the time of day. "Cancel" and "OK" buttons are provided to either discard or confirm the selected time.

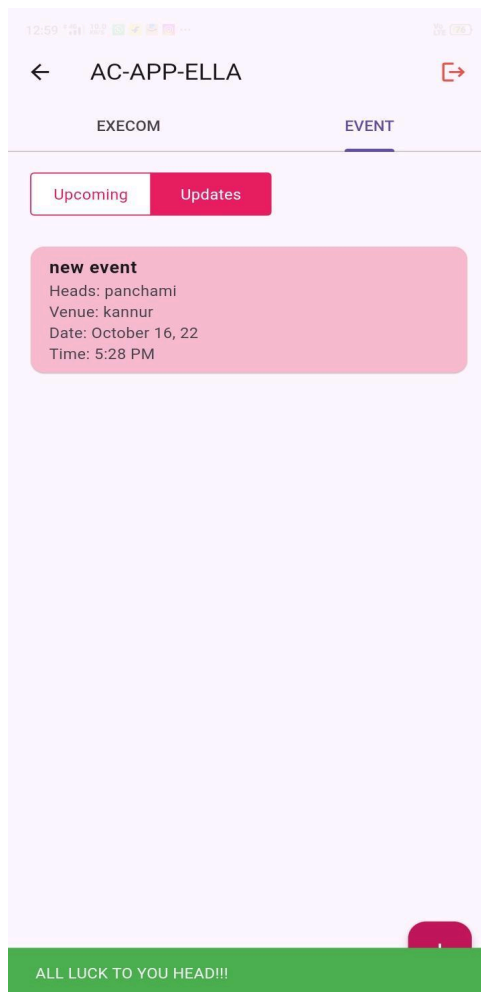


Fig 5.15 Committee Home Page -
Update Tab

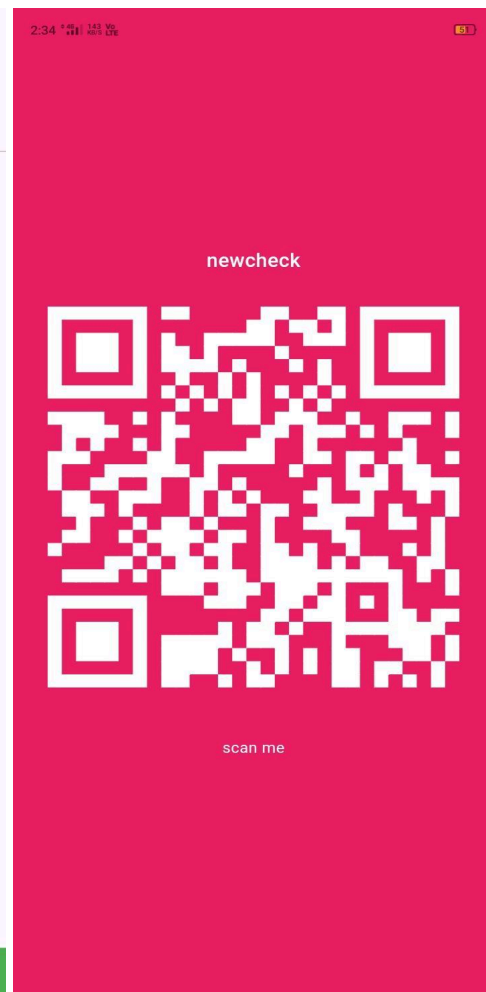


Fig 5.16 QR Screen

- This page, titled "Upcoming" under the "EVENT" section, serves to display a list of all the events that are scheduled to happen in the future for the club. It provides a quick and easy way for club members to see what events are coming up, allowing them to plan their participation accordingly. Each listed event offers essential details to inform members about what's on the horizon.

- This Flutter screen displays a prominent QR code against a pink background, associated with the event titled "newcheck." Below the QR code, the instruction "scan me" indicates that this visual code is intended to be scanned, likely for purposes such as event check-in or attendance tracking within the AC-APP-ELLA application.

CHAPTER 6

CONCLUSION

The AC-APP-ELLA application provides an efficient and structured solution for college club management by streamlining event organization and attendance tracking. The implementation of QR code-based attendance ensures accurate and hassle-free participation tracking, reducing manual errors. The Firebase Firestore database enables real-time data synchronization, ensuring users have up-to-date event information. A username-password authentication system allows secure access, making the app simple and user-friendly. The dashboard offers a centralized view of events, helping Committee Members and Club Members efficiently manage and participate in club activities. With its structured design and seamless functionality, AC-APP-ELLA enhances club operations by improving efficiency and user engagement. Overall, the application simplifies club management while providing a scalable and reliable platform. Future improvements can further enhance its features, making it an essential tool for managing college club activities effectively.

6.1 FUTURE SCOPE

The AC-APP-ELLA application has significant potential for future enhancements to improve functionality, user experience, and efficiency. Advanced event analytics can be integrated to provide insights into participation trends and engagement levels, helping organizers make informed decisions. Enhancing user roles and permissions will allow better role-based access control, ensuring a more organized management system. Implementing offline mode support will enable users to access event details and mark attendance without an internet connection, with data syncing once online. Expanding the application to web and iOS platforms will enhance accessibility and usability across different devices. Personalized event recommendations based on user interests and past participation can increase engagement. Additionally, incorporating an in-app communication system will facilitate better interaction between committee members, organizers, and club members. An automated event certificate generation feature can also be introduced to streamline the distribution of participation certificates. By integrating these enhancements, AC-APP-ELLA can evolve into a more comprehensive and scalable platform for efficient college club management.

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